RSCS 1.0 Foundations:
A Typed Coordinate and Operator Framework
as a Conservative Extension of RGCS v2

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Abstract

The Resonant Spacetime Coordinate System (RSCS 1.0) is a typed mathematical layer above the frozen RGCS v2.0.0 crystal-application core: 17 coordinate types on declared manifolds and 23 operators with units, classification, source provenance, and machine tests. Its defining discipline is the *Conservative Extension Property*: for every operator that generalizes a frozen v2 result, $O_{\text{RSCS}}(\iota(x)) = \iota(O_{\text{RGCS}}(x))$ within declared tolerance on the v2 domain, so the extension can never silently change v2 (battery status: all pass). Classification tags (**[EST]** established, **[DER]** derived, **[HYP]** hypothesis, **[SRC]** source claim, **[ENG]** engineering) attach to every definition, and a machine-enforced firewall forbids strong claims built from weak inputs. RSCS provides models and falsification protocols; it asserts no unconfirmed physics.

1 Introduction

RGCS v2 established a deterministic, tested crystal-resonance core with strict claim classification. Version 3 asks a structural question: can the same discipline support a broader family of coordinates (phase, polarization, orientation, group delay, provenance) and operators (coupling, propagation, preparation, observation) without weakening the frozen baseline? RSCS answers with three rules:

1. **Typed states.** Every state is an immutable coordinate record on a declared manifold ($\mathbb{E}^3$, S^1, $SO(3)$, $\mathbb{C}^n$, ...) validated at construction; flat untyped vectors are non-compliant (D3-009).
2. **Conservative extension.** New operators must reproduce the frozen results they generalize (Section 5).
3. **Provenance as data.** Classification and source identity travel with the state; a machine firewall (D3-012) rejects any **[EST]**/**[DER]** output computed from **[HYP]**/**[ENG]**/**[SRC]** inputs without an explicit downgrade.

The framework *earns its complexity* case by case: wherever a simpler model predicts equally well, the registry says so and the simple model wins (v2 rule, inherited).

2 Architecture

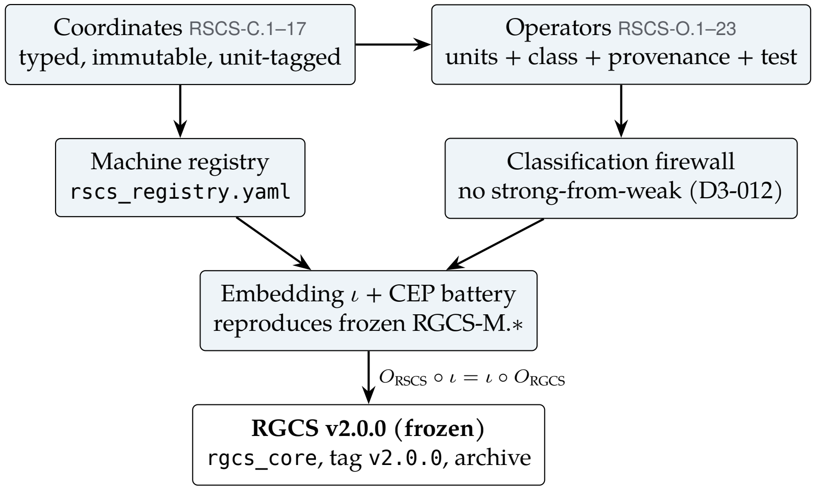

Figure 1: State/operator architecture. The frozen v2 core is never modified; RSCS reaches it only through the embedding, and the CEP battery pins the round trip.

3 Coordinates

Table 1: RSCS coordinates (machine registry `rscs_registry.yaml`); RSCS-C.15 is defined in the memory application layer.

ID	Name	Class	Units	Manifold
RSCS-C.1	spatial coordinate	EST	mm (x, y, z) in E^3	E^3
RSCS-C.2	time	EST	s	R
RSCS-C.3	phase coordinate	EST	rad (wrapped to [0, 2*pi))	S^1
RSCS-C.4	angular frequency	EST	rad/s (omega = 2*pi*f)	R
RSCS-C.5	wavevector / oriented frame element	EST	rad/mm in R^3	R^3
RSCS-C.6	mode index tuple	EST	integer indices	Z^n
RSCS-C.7	modal state (occupancy)	EST	complex amplitudes (campaign observable X)	C^n
RSCS-C.8	orientation / reference frame	EST	SO(3) x +/-	SO(3) x +/-
RSCS-C.9	polarization / spin state	EST	Stokes 3-vector on S^2 (dimensionless)	S^2
RSCS-C.10	selection coordinate	DER	class label (selector unit) + population [0,1]	R or Z
RSCS-C.11	group-delay coordinate	DER	s (per mode)	R^n
RSCS-C.12	uncertainty descriptor	DER	(value in caller unit, relative 1-sigma, dist)	-
RSCS-C.13	provenance tag	EST	(source_id, class, path)	enum tuple
RSCS-C.14	memory-lattice coordinate	HYP	(orientation index, carrier phase rad) on T^n	T^n
RSCS-C.15	Hydrogenuine memory record	ENG	engineering record (composite of C.1/C.8/C.2/C.3/C.7/C.12/C.13)	product
RSCS-C.16	optical carrier	EST	wavelength nm; envelope bandwidth Hz	R+ x R+

ID	Name	Class	Units	Manifold
RSCS-C.17	directional propagation pair	DER	rad/mm (beta, delta_beta)	R^2

Coordinates reject non-finite components at construction, wrap periodic quantities onto S^1, and normalize where the manifold demands it (the polarization state RSCS-C.9 lives on the Poincaré sphere and carries exact Jones↔Stokes conversions). The memory-lattice coordinate RSCS-C.14 is **[HYP]** and quarantined: it cannot be constructed without an explicit `acknowledge_hypothesis` flag (D3-013).

4 Operators

Table 2: RSCS operators; the last column names the frozen v2 results each operator must reproduce (machine-tested).

ID	Name	Class	Reproduces
RSCS-O.1	frame transform	EST	–
RSCS-O.2	time<->frequency map (analytic signal)	EST	RGCS-M.55
RSCS-O.3	space->phase map	HYP	–
RSCS-O.4	mode-coupling operator (anti-Hermitian)	EST	RGCS-M.23, RGCS-M.24, RGCS-M.28, RGCS-M.46
RSCS-O.5	parity / supermode basis change	EST	RGCS-M.23, RGCS-M.24, RGCS-M.25
RSCS-O.6	transfer-matrix cascade	EST	–
RSCS-O.7	phase-matching predicate	EST	–
RSCS-O.8	group-delay / dispersion balance	DER	–
RSCS-O.9	state preparation	DER	–
RSCS-O.10	observation / metric	EST	RGCS-M.56
RSCS-O.11	uncertainty propagation	DER	RGCS-M.10, RGCS-M.11
RSCS-O.12	provenance propagation	EST	–
RSCS-O.13	memory-lattice store / recall	HYP	–
RSCS-O.14	HG memory store (write)	ENG	–
RSCS-O.15	HG memory replay (recall)	ENG	–
RSCS-O.16	HG memory update (predict<->observe)	ENG	–
RSCS-O.17	anisotropic elastic wave speeds (Christoffel)	EST	RGCS-M.10
RSCS-O.18	dispersion phase expansion	EST	–
RSCS-O.19	photon-phonon mode conversion (selection rules)	DER	–
RSCS-O.20	Autler-Townes split response	EST	RGCS-M.24
RSCS-O.21	critical coupling	EST	–
RSCS-O.22	state-dependent susceptibility + nonreciprocal metrics	DER	–
RSCS-O.23	directional propagation split + beating length	EST	–

4.1 The coupling keystone RSCS-O.4

The time-domain coupling of a degenerate mode pair is anti-Hermitian,

$$K = i\,2\pi g, \qquad K^\dagger = -K, \qquad \text{RSCS-O.4 [EST]} \tag{1}$$

inheriting the v2 QA correction QA-D-04 and frozen by D3-007. The frequency-domain Hermitian matrix $H = \mathrm{diag}(f) + g$ yields hybrid frequencies $f_0 \pm g$: for the golden case ($f_A = f_B = 1000\,\mathrm{Hz}$, $g = 10\,\mathrm{Hz}$) the operators return 990.0 and 1010.0 Hz, a splitting of 20.0 Hz ($= 2g$), matching frozen RGCS-M.24 exactly. The forbidden real-symmetric $K = \pi g$ grows without bound and is pinned as a contrast regression test. The Autler–Townes dressed response RSCS-O.20 **[EST]** maps onto the same algebra with $G = 2\pi(2g)$ [1]; Figure 2 shows the weak/strong regimes.

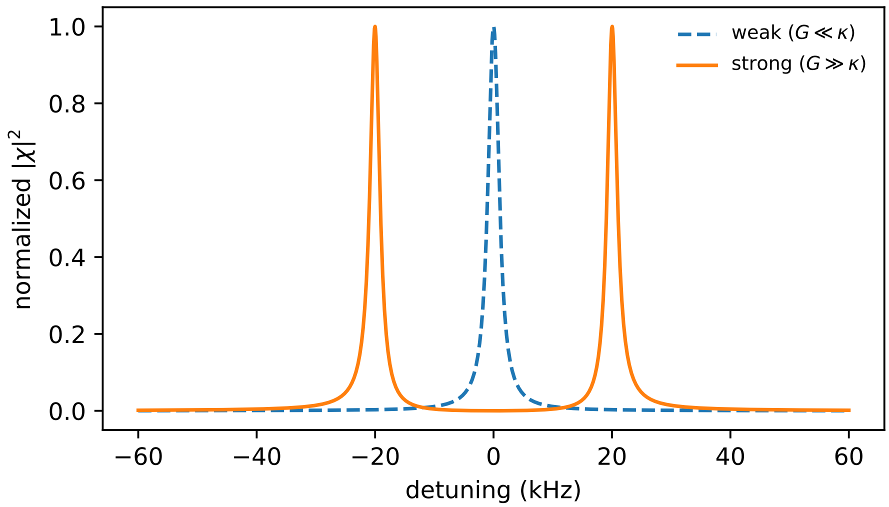

Figure 2: Two-Lorentzian dressed lineshape RSCS-O.20: a single peak for $G \ll \kappa$, a resolved doublet split by G for $G \gg \kappa$ (threshold $G > \sqrt{\kappa_1 \kappa_2}$). Lineshape algebra only; no device physics is imported **[SRC]**.

4.2 Adapted mathematics and the firewall

Operators O.6–O.9 and O.18–O.23 adapt published mathematics (transfer-matrix cascades, phase matching, group-delay balancing, state-dependent susceptibility, directional propagation) [2–5]. Every adaptation is pinned to an equation-provenance row (EP-$*$) with a *binding* forbidden-transfer clause; the reciprocity posture is explicit: an unbiased passive quartz path is reciprocal, and the pre-registered expectation for every directional observable is the null (D6-003). The Hydrogenuine memory record RSCS-C.15 and its store/replay/update operators RSCS-O.14–16 are **[ENG]** software abstractions; the NHT space-to-phase encoding RSCS-O.3 and HAL lattice remain **[HYP]** and quarantined [6, 7].

5 The Conservative Extension Property

The embedding ι maps every v2 state into its typed RSCS counterpart. For each generalizing operator the battery checks $O_{\mathrm{RSCS}}(\iota(x)) = \iota(O_{\mathrm{RGCS}}(x))$ with rtol 10^{-9} / atol 10^{-12} (D3-011) across the frozen results RGCS-M.23/24/28/46/55/56 and 10–11:

Table 3: Conservative Extension Property battery (`rscs_core.embedding.run_all_cep`); tolerances rtol 10^{-9}, atol 10^{-12} (D3-011).

Check	Result
`two_mode`	pass
`two_mode`	pass
`n_mode`	pass
`analytic_signal`	pass
`coherence`	pass
`uncertainty`	pass

The battery runs in the test suite on every commit; a CEP failure is a release blocker, not a warning.

6 Identifiability, gauge, and uncertainty boundary

Phase coordinates carry an S^1 gauge (a global phase is unobservable); parity/supermode bases RSCS-O.5 are unitary and self-inverse; frame transforms RSCS-O.1 declare their singularity handling. Uncertainty RSCS-C.12 is a coordinate, propagated by RSCS-O.11 **[DER]** in agreement with the v2 propagation rules; no RSCS operator may output a value stripped of its uncertainty when the input carried one.

7 Status and scope

RSCS 1.0 is software and mathematics with machine evidence; it confirms no new physics. Its hypotheses (H-20 – H-30 in the claim register) are pre-registered with observables, controls, and failure conditions — and several are pre-registered *nulls*. The framework's value claim is reproducibility: every number in this document was generated from the tested libraries at build time.

References

[1] Donggyu B. Sohn, Oğulcan E. Örsel, and Gaurav Bahl. Electrically driven linear optical isolation through phonon mediated Autler-Townes splitting, 2021. Department of Mechanical Science and Engineering, University of Illinois at Urbana–Champaign.

[2] Haotian Cheng, Yishu Zhou, Freek Ruesink, Margaret Pavlovich, Shai Gertler, Andrew L. Starbuck, Andrew J. Leenheer, Andrew T. Pomerene, Douglas C. Trotter, Christina Dallo, Matthew Boady, Katherine M. Musick, Michael Gehl, Ashok Kodigala, Matt Eichenfield, Anthony L. Lentine, Nils T. Otterstrom, and Peter T. Rakich. A terahertz bandwidth nonmagnetic isolator, 2024.

[3] Zhu-Bo Wang, Yan-Lei Zhang, Xin-Xin Hu, Guang-Jie Chen, Ming Li, Peng-Fei Yang, Xu-Bo Zou, Peng-Fei Zhang, Chun-Hua Dong, Gang Li, Tian-Cai Zhang, Guang-Can Guo, and Chang-Ling Zou. Self-induced optical non-reciprocity. *Light: Science & Applications*, 14:23, 2025. doi: 10.1038/s41377-024-01692-y.

[4] Shicheng Zhang, Yifan Zhan, Shangqing Gong, and Yueping Niu. Noiseless single-photon isolator at room temperature. *Communications Physics*, 2023. doi: 10.1038/s42005-023-01141-z.

[5] Kimhong Chao, Vy Yam, Laurent Vivien, and Béatrice Dagens. Integrated TE optical isolator based on magneto-optical perturbation in coupled waveguides. *Scientific Reports*, 2025. doi: 10.1038/s41598-025-08507-z.

[6] Katsushi Arisaka. Grand unified theory of mind and brain, part II: Neural holographic tomography (NHT) and holographic ring attractor lattice (HAL). V1 manuscript, 2022-03-05. University of California, Los Angeles. SRC-class source; not peer-reviewed, 2022.

[7] Katsushi Arisaka and Aaron Blaisdell. Grand unified theory of mind and brain, part III: Holographic visual perception of 3D space and shapes. V1 manuscript, 2022-03-05. University of California, Los Angeles. SRC-class source; not peer-reviewed, 2022.